

211E - 211ER



Control board 230V for automation of roller shutters
Instructions and warnings

ENGLISH

INDEX

OVERVIEW	6
1 PRODUCT CONFORMITY	6
2 WARNINGS SUMMARY	6
3 MODELS AND CONTENTS OF THE PACKAGE	7
4 PRODUCT DESCRIPTION	7
5 TECHNICAL DATA	7
6 FORESEEN OPERATING CONDITIONS	7
7 ASSEMBLY AND WIRING INSTRUCTIONS	8
8 OPERATING INSTRUCTIONS	8
8.1 Learning of working times	8
8.2 Description of functioning program	8
8.3 Radio receiver	9
9 MAINTENANCE	9
10 PRODUCT DISPOSAL	10
11 COMPLETE CLOSING ASSEMBLY	10

OVERVIEW

These instructions were prepared by the manufacturer and are an integral part of the product. The operation described are designed for adequately trained and qualified personnel and must be carefully read and kept for future reference.

1 PRODUCT CONFORMITY

The 211E - 211ER programmable control board bears the CE label. DEA SYSTEM guarantees the conformity of the product to European Directives 2004/108/CE (concerning electromagnetic compatibility), 2006/95/CE (low voltage electrical equipment).

2 WARNINGS

Read these warnings carefully. Failure to respect the following warnings may cause risk situations.

WARNING DEA System reminds all users that the selection, positioning and installation of all materials and devices which make up the complete automation system, must comply with the European Directives 2006/42/CE (Machinery Directive), 2004/108/CE (electromagnetic compatibility), 2006/95/CE (low voltage electrical equipment). In order to ensure a suitable level of safety, besides complying with local regulations, it is advisable to comply also with the above mentioned Directives in all extra European countries. A1

WARNING Using the product under unusual conditions not foreseen by manufacturer may cause dangerous situations; this is the reason why all the conditions prescribed in these instructions must be followed. A2

WARNING Under no circumstance must the product be used in an explosive environment or surroundings that may prove corrosive and damage parts of the product. A3



⚠ WARNING To ensure an appropriate level of electrical safety always keep the 230V power supply cables apart (minimum 4 mm in the air or 1 mm through insulation) from low voltage cables (motors power supply, controls and aerial circuits power supply) and fasten the latter with appropriate clamps near the terminal boards . A4

⚠ WARNING The device shouldn't be used by children and/or persons with reduced capacities (physical, sensorial, mental). Prevent children from playing with the device. All these operations must be performed only after disconnecting the power supply and operating in strict compliance with the electrical standards and regulations in force in the nation of installation. A5

⚠ WARNING For further details and control board installation and wire make reference to the instructions given (such as hole drilling to allow for wires passage, use of wire clamps..). Failure to comply with these instructions may jeopardize the level of electrical safety. A6

⚠ WARNING The 211E - 211ER control panel hasn't any internal devices for the protection of mechanical risks following to the gate/door movement (es: limitation of impact forces). If such risks are founded, external protection devices (independent from the control panel) are necessary. An erroneous evaluation of impact forces may cause serious damages to persons, animals, things. DEA System reminds the installer to check these impact forces, measured following the EN 12445 standard. Such impact forces must be lower than what foreseen from the EN 12453.

⚠ WARNING Other eventual protection devices to respect the impact forces must be in conformity with EN 12978 standard. A9

⚠ WARNING Using spare parts not indicated by DEA System and/or incorrect re-assembly may endanger people, animals and property, and may also cause malfunctioning of the products: always use parts provided by DEA System and follow assembly instructions. A10

⚠ WARNING Disposal of packing materials (such as plastic, card board, etc..) must be done according to regulations in force locally. Do not leave plastic bags and polystyrene within the reach of children. A11

3 MODELS AND CONTENTS OF THE PACKAGE

The 211E - 211ER control board is supplied in a box which permits the external assembly with IP 54 protection degree.

4 PRODUCT DESCRIPTION

The control boards 211E and 211ER have been designed and tested exclusively for the automation of roller shutters with a DEA SYSTEM 230V ~ operator. Their main features are:

- easy installation;
- 3 different operating programs (step by step – step by step with separate open/close command – present man command);
- (only art. 211ER): built-in 433.92 MHz radio receiver for both HCS or dip switch coding.

5 TECHNICAL DATA

Power supply	230 V ~ - 50Hz
Max motor capacity	500W max
Fuse F1	T5A 250V retarded
Radio receiver frequency	433,92 MHz rolling code/dipswitch code
Max. number of radio controls managed	40
Protection level	IP 54

6 OPERATING CONDITIONS

211E - 211ER control board is designed for the automation of swing gates operated by 230 V ~ motors. This control board has been designed and tested for operation under "normal" conditions for both residential use.



⚠ WARNING Using the product under unusual conditions not foreseen by manufacturer may cause dangerous situations; this is the reason why all the conditions prescribed in these instructions must be followed. A2

⚠ WARNING Under no circumstance must the product be used in an explosive environment or surroundings that may prove corrosive and damage parts of the product. A3

7 ASSEMBLY AND WIRING INSTRUCTIONS

Connect to the power supply 230 V ~ 50 Hz through a multi pole switch or a different device that can ensure multi pole disconnection from the power supply, with a contact opening of 3,5 mm. Use a cable with a minimum section of 3 x 1,5 mm² (e.g. a H07RN-F type).

Table 1 terminal boards connection

1-2		230V ~ 50Hz power supply input (L=phase, N=neutral)
3-4-5		230V ~ 50Hz motor output max 500W (3=open, 4=common, 5=close)
6		Input for N.O. closing push-button or N.C. photocell contact (this input may be used in different way depending on the selected program. See par. 8.1) WARNING: the control panel doesn't foresee an exit for auxiliary power supply. The eventual photocell has to be separately powered.
7		Input for opening push-button
8	COM	Common inputs
9		Antenna signal (only for art. 211ER)
10		Antenna ground (only for art. 211ER)

8 OPERATING INSTRUCTIONS

8.1 Description of operating programs

This control panel contains 3 different operating programs and a programming mode to memorize the operating time; these programs can be selected by dip-switch no.1 as indicated in table 2.

WARNING: once the operating time is selected, cut off and re-apply power supply.

Table 2 operating programs

DIP1	DIP2	PROGRAMS
OFF	OFF	MAN PRESENCE: open command maintained pushed→opening, close command maintained pushed→closing. Door stops at command release
ON	OFF	STEP BY STEP with SEPARATED OPEN/CLOSE: open impulse→opening, close→stop, close→opening, open→stop, open→opening... If during the opening or closing cycle, any open or close command is activated, the operator stops at working time expire. No automatic closing foreseen.
OFF	ON	STEP BY STEP with PHOTOCCELL: open impulse→opening, open→stop, open→closing, open→stop, open→opening... If during the closing cycle the close input is activated (photocell contact) the door reverses its run. If during the opening or closing any open impulse or photocell are activated, the operator stops at working time expire. No automatic closing foreseen. WARNING: remember to bridge CLOSE input (Photocell) towards the common if not in use. WARNING: the control panel doesn't foresee an exit for auxiliary power supply. The eventual photocell has to be separately powered.



ON	ON	PROGRAMMING: control panel waits for commands for the operating time memorizing procedure. See par. 8.2. Warning: you may get access to programming only if control panel is at feed or at closed, not during a working cycle.
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8.2 Operating times learning

1. Execute wirings to terminals, control panel must not be at feed, manually place the door at half run;
2. Set DIP1=ON and DIP2=ON and apply power supply: the flashing light flickers: the control panel is ready for the programming;
3. Give a START input: (by push-button panel or remote) door starts opening;
4. Once the door reaches the point of complete opening, give another OPEN impulse: the door stops;
5. Give another START input: door starts again to close;
6. Once the door reaches the point of complete opening, give another OPEN impulse: the door stops and the flashing light is fixed;
7. Adjust the dip switches as desired according to the operating program and exit the programming (see Table 2): the LED is switched off and the learning of operating times is finished. The following cycle will be executed according to the selected operating program.

8.3 Radio receiver

211ER control board includes a 433,92MHz built-in radio receiver accepting both transmitters with HCS coding (complete rolling code or just part fixed), and HT12E dip-switch coding. The receiver memory capacity can contain up to **40** different transmitters.

It is possible to choose, according to one's needs, which key of the memorized transmitters will activate the open input and which one will activate the close input. Procedures for the memorizing and deleting of the remotes are explained as follows.

Memorization of transmitters:

1. Cut off and then apply power to the control panel or ensure it has finished a closing cycle and it's ready to receive instructions;
2. Press S2 key on the control panel and continue pressing it (by keeping it pressed the receiver is learning);
3. Press the key you want to link to OPEN input: flashing light blinks (in case you don't want to link any channel to open input, go directly to point 5);
4. Press the key you want to link to CLOSE input: flashing light blinks (in case you don't want to link any channel to close input, go directly to point 5);
5. Repeat points "3" and "4" for other remotes by keeping S2 pressed;
6. Release S2 key: learning completed.

WARNING: if the S2 key is maintained pressed for 30 seconds without any remote being learnt, a reset procedure of the remote memory is started up!! See "Deletion of transmitters".

Deletion of transmitters:

1. Cut off and apply power to the control panel or ensure it has completed a closing cycle and it's ready to receive instructions;
2. Press S2 key on the control panel and continue pressing it: after 30 seconds the flashing light turns on.

WARNING: during this period, ensure that any remote doesn't transmit.

2. Keep S2 key pressed: after 15 seconds the flashing light blinks;
3. Release the key S2: memory cancelled.



9 MAINTENANCE

 **WARNING** Any installation, maintenance or repair operation on the whole system must be carried out exclusively by qualified personnel. All these operations must be performed only after disconnecting the power supply, and operating in strict compliance with the electrical standards and regulations in force in the nation of installation.

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10 PRODUCT DISPOSAL



In line with EU Directive 2002/96/EC for waste electrical and electronic equipment (WEEE), this electrical product must not be disposed of as unsorted municipal waste. Please dispose of this product by returning it to your local municipal collection point for recycling.

11 COMPLETE CLOSING ASSEMBLY

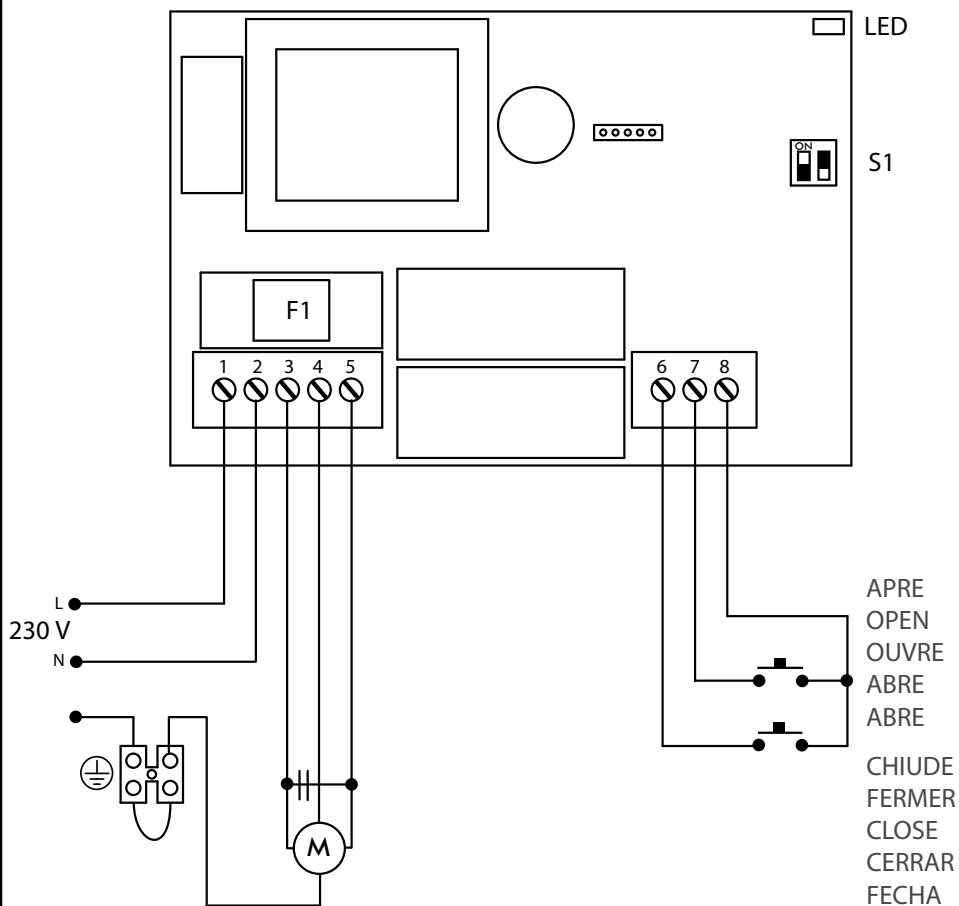
Remember that everyone who sells and/ or automates doors/gates becomes the manufacturer of the automated door/gate machine, and must therefore prepare and preserve a technical folder containing the following documents (see Machinery Directives Enclosure V).

- Assembly drawing of the automated door/gate.
- Electrical connection and control circuit diagram.
- Risk analysis including: a list of the essential safety requirements provided in machine Directive Enclosure I; a list of the risks posed by the door/gate and a description of the implemented solutions
The installer must also:
- Keep these operating instructions and the instructions for all other components in a safe place.
- Prepare the operating instructions and general safety warnings (by filling up these operating instructions) and hand a copy to the end user.
- Fill in the maintenance handbook and hand a copy to the end user.
- Draw up the EC declaration of conformity and hand a copy to the end user.
- Fill in the complete EC label or plate and apply it to the automated door/gate.

N.B. The technical folder must be kept for inspection by the competent national authorities for at least ten years starting from the date of manufacturing of the door/gate operator.

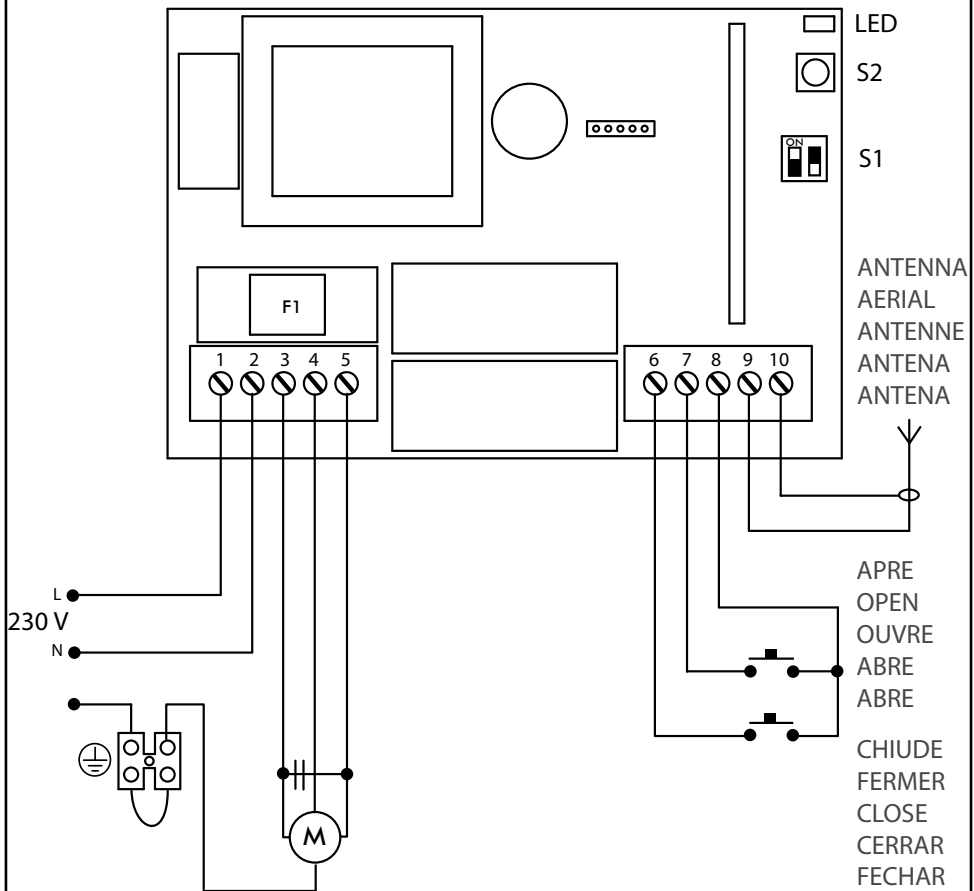


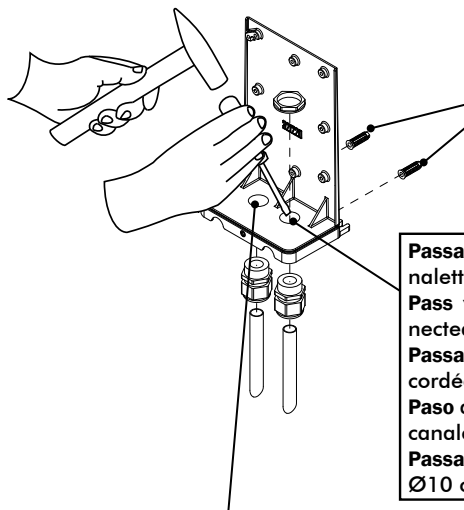
Schema elettrico con 211E, 211E electrical diagram, Schéma électrique avec 211E, Esquema eléctrico con 211E, Esquema eléctrico para 211E.





Schema elettrico con 211ER, 211ER electrical diagram, Schéma électrique avec 211ER, Esquema eléctrico con 211ER, Esquema eléctrico para 211ER.





Eseguire il fissaggio alla parete usando opportuni tasselli per viti Ø5 (non fornite)

Fix the box on the wall with appropriate bushings to anchor screws Ø5 (not included)

Le fixer au mur en utilisant des douilles à expansion pour vis adéquates Ø5 (pas incluses)

Efectuar la fijación a la pared utilizando adecuados tacos para tornillos de Ø5 (no incluidos)

Executar a fixação a parede usando apropriadas rolhas para parafusos Ø5 (não fornecidas)

Passaggio cavi a bassissima tensione all'interno di una canaletta Ø10 raccordata con fermatubo PG16 (non forniti)

Pass very low tension cables inside a grommet Ø10 connected with a tube fastening PG16 (items not included)

Passage des fils à très basse tension dans un passe-fil Ø10 raccordée avec un bloque tube PG16 (ces outils ne sont pas inclus)

Paso de los cables de tensión muy baja por el interior de una canaletta de Ø10 unida con paratubo PG16 (no incluidos)

Passagem cabos a baixíssima tensão ao interno de um cano Ø10 com fixação do tubo PG16 (não fornecidos)

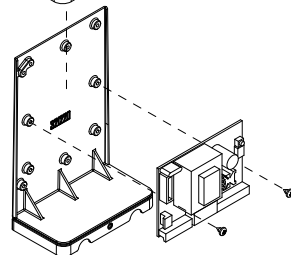
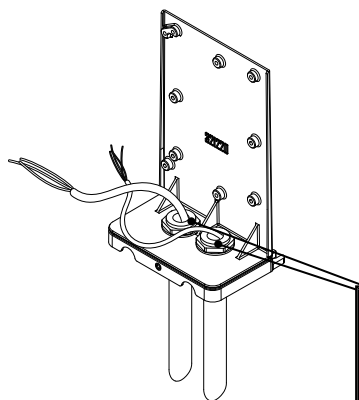
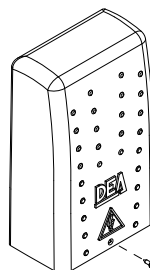
Passaggio cavi 230V all'interno di una canaletta Ø10 raccordata con fermacavi PG16 (non forniti)

Pass 230V cables inside a grommet Ø10 connected with a tube fastening PG16 (items not included)

Passage des fils 230V dans un passe-fil Ø10 raccordée avec un bloque tube PG16 (ces outils ne sont pas inclus)

Paso de los cables 230V por el interior de una canaletta de Ø10 unida con paratubo PG16 (no incluidos)

Passagem cabos 230V ao interno de um cano Ø10 com fixação do tubo PG16 (não fornecidos)



Sigillare le canalette dopo il passaggio dei cavi

Seal the tubing trays after installing the wires

Étanchez les passe-fils après que vous avez passé des fils

Una vez colocados los cables, tapar las canaletas

Tapar os cabos depois de passar os fios eléctricos